# Problem Set 5

1. Random Sample: $\bar{X}=52$, $s=10$, $n=100$

Compute SE of $\bar{X}$. $H_0: \mu=50$, vs. $H_1: \mu \neq 50$

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$$

$$z = \frac{\bar{X} - \mu_0}{\widehat{SE}} = \frac{52-50}{1} = 2$$

| $\alpha$ | Confidence | Critical Val | $\lvert z \rvert >$ Critical Val | |
|---|---|---|---|---|
| 0.1 | 90% | 1.645 | Yes | Reject the null $H_0$ |
| 0.05 | 95% | 1.96 | Yes | Reject the null $H_0$ |
| 0.01 | 99% | 2.576 | No | Fail to Reject the null $H_0$ |

95% CI:

$$\bar{X} \pm 1.96 \cdot \widehat{SE} = 52 \pm 1.96 \cdot 1 = (50.04, 53.96)$$

2. $\hat{S}=3.1$, $\widehat{SE}(\hat{S})=1.3$, $H_0: S=0$, $H_1: S \neq 0$

$$z = \frac{\hat{S}-0}{\widehat{SE}} = \frac{3.1}{1.3} \approx 2.38$$

| | | CI | Contains 0 | |
|---|---|---|---|---|
| 90% | $3.1 \pm 1.645 \cdot 1.3$ | (0.96, 5.24) | No | Reject $H_0$ |
| 95% | $3.1 \pm 1.96 \cdot 1.3$ | (0.55, 5.65 | No | Reject $H_0$ |
| 99% | $3.1 \pm 2.576 \cdot 1.3$ | (-0.75, 6.45) | Yes | Fail to Reject $H_0$ |

3. $\bar{X} = 11.5$ , $S = 3$ , $n = 36$ , $H_0: \mu = 12$, $H_1: \mu \neq 12$

$$\widehat{SE} = \frac{3}{\sqrt{36}} = \frac{3}{6} = 0.5$$

$$Z = \frac{11.5 - 12}{0.5} = \frac{-0.5}{0.5} = -1$$

$11.5 \pm 1.96 \cdot 0.5 = 11.5 \pm 0.98 = (10.52, 12.48)$

12 is in the CI, fail to reject $H_0$

4. $\bar{X}_A = 25$ , $S_A = 8$ , $n_A = 100$ , $H_0: \mu_A - \mu_B = 0$

$\bar{X}_B = 22$ , $S_B = 7$ , $n_B = 100$ $H_1: \mu_A - \mu_B \neq 0$

$\bar{X}_A - \bar{X}_B = 25 - 22 = 3$

$$\widehat{SE}(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}} = \sqrt{\frac{64}{100} + \frac{49}{100}} = \sqrt{1.13} = 1.063$$

$$Z = \frac{3 - 0}{1.063} = 2.82$$

$3 \pm 1.96 \cdot 1.063 = (0.92, 5.08)$

0 not in CI, reject $H_0$

5. $\bar{X}_A = 14$, $S_A = 6$, $n_A = 81$ $H_0: \mu_A - \mu_B = 0$

$\bar{X}_B = 12$, $S_B = 5$, $n_B = 64$ $H_1: \mu_A - \mu_B \neq 0$

$\bar{X}_A - \bar{X}_B = 14 - 12 = 2$

$\frac{S_A^2}{n_A} = \frac{36}{81} = 0.444$ $\frac{S_B^2}{n_B} = \frac{25}{64} = 0.391$

$\widehat{SE} = \sqrt{0.444 + 0.391} = \sqrt{0.835} = 0.914$

$z = \frac{2 - 0}{0.914} = 2.19$

6. $\bar{X}_A = 3.2$, $S_A = 1.5$, $n_A = 49$ $H_0: \mu_A - \mu_B = 0$

$\bar{X}_B = 2.8$, $S_B = 1.4$, $n_B = 64$ $H_1: \mu_A - \mu_B \neq 0$

$\bar{X}_A - \bar{X}_B = 3.2 - 2.8 = 0.4$

$\frac{S_A^2}{n_A} = \frac{2.25}{49} = 0.0459$ $\frac{S_B^2}{n_B} = \frac{1.96}{64} = 0.0306$

$\widehat{SE} = \sqrt{0.0459 + 0.0306} = \sqrt{0.0765} = 0.277$

$z = \frac{0.4}{0.277} = 1.45$

$0.4 \pm 1.96 \cdot 0.277 = (-0.14, 0.94)$

0 in the CI, Fail to reject $H_0$